

Experiment No - 1

Objective -

- To prove a male-female V-joint by hand fitting operations, as per the given drawing.
- To prepare Cross Half-Lap joint, Half Lap Dovetail joint & Mortise & Tenon joint in wood, as per given dimension.

Apparatus / Equipment Required.

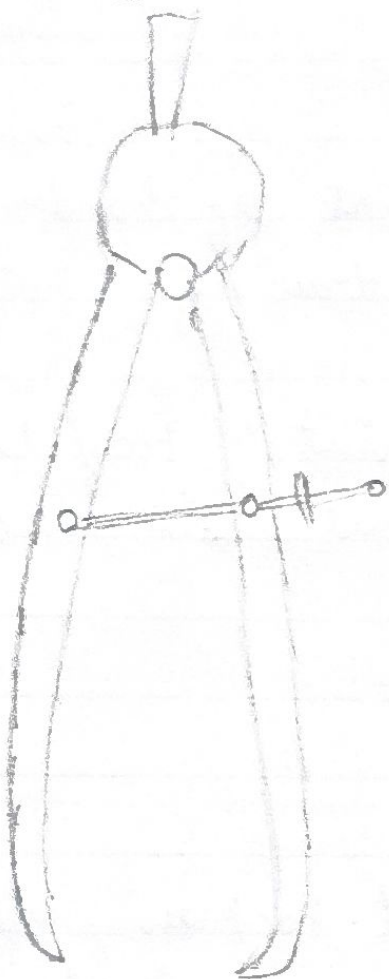
Fitting Shop: Bench vice, hand hacksaw, double-cut file, single-cut file, triangular file, centre punch, hammer.

Carpentry Shop: Carpentry vice, steel rule, try square, iron jack plane, marking gauge, tenon saw, rip saw, mallet, pencil, sandpaper, wooden pieces (150mm x 75mm x 25mm) 2 nos.

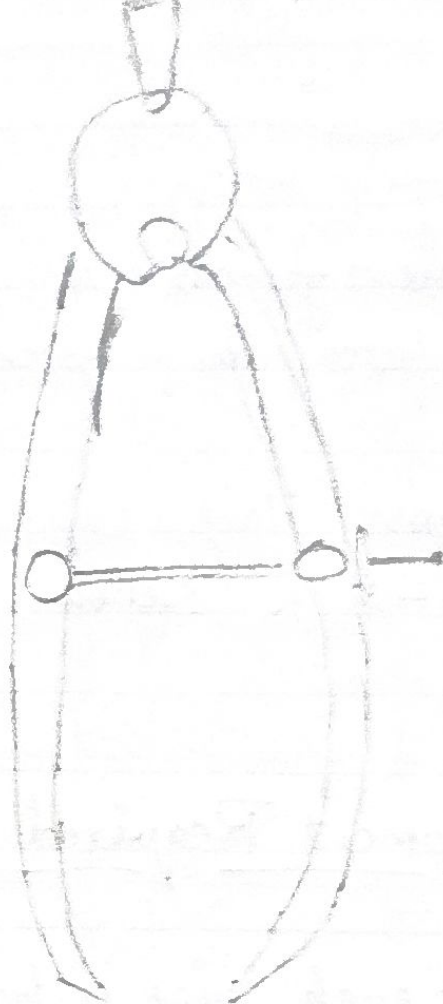
Scriber



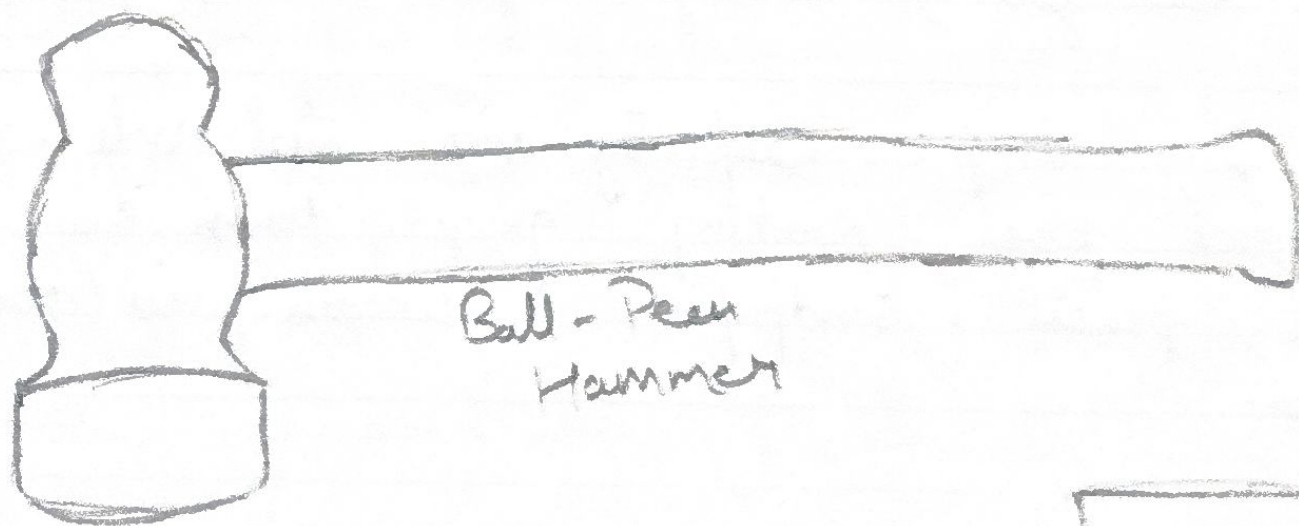
Outside
Calliper



Inside
Calliper



Centre
Punch



Ball - Peen
Hammer



Fig
50/20

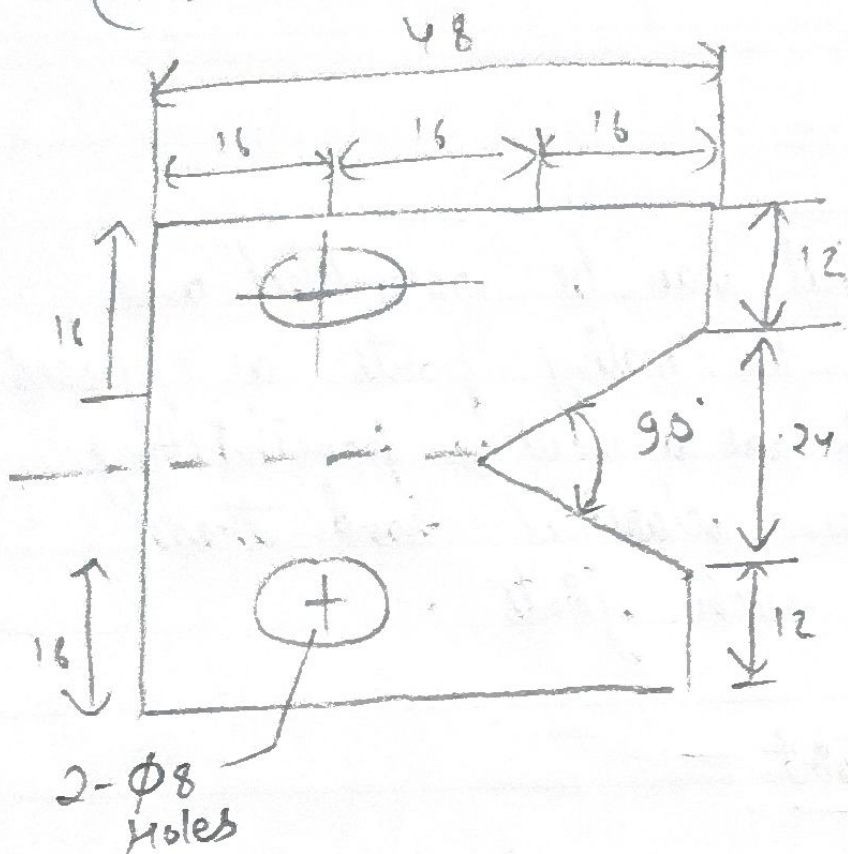
Theory -

A temporary joint is one that can be assembled and dismantled without damaging the mating parts as opposed to a permanent joint such as a weld. Fitting and carpentry are the two classical hand trade processes used to make such joints.

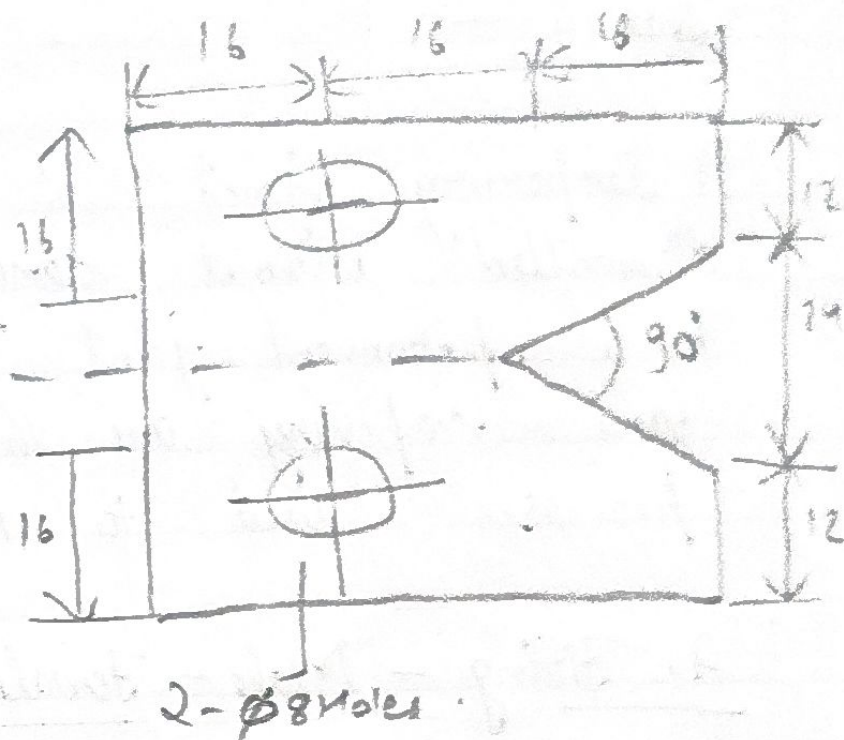
A. Fitting - Male - Female V-Joint -

Fitting is the process of finishing / matching two or more machined or hand-worked parts to assemble them accurately. In a male-female V-joint, the male part carries a projected V-shaped portion, and the female part has a recess of the same dimension into which the male part fits, forming an interference free sliding joint.

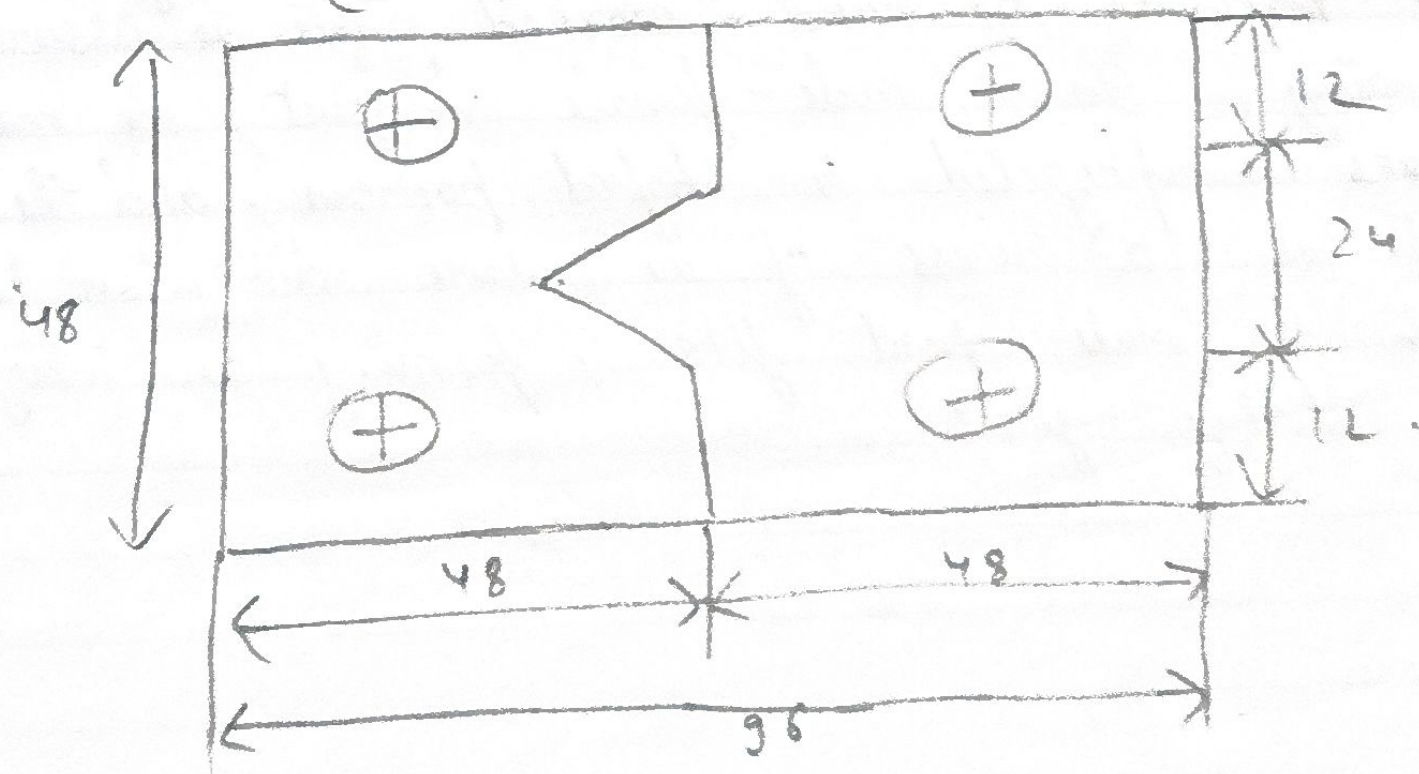
(A) Male Part



(B) Female Part

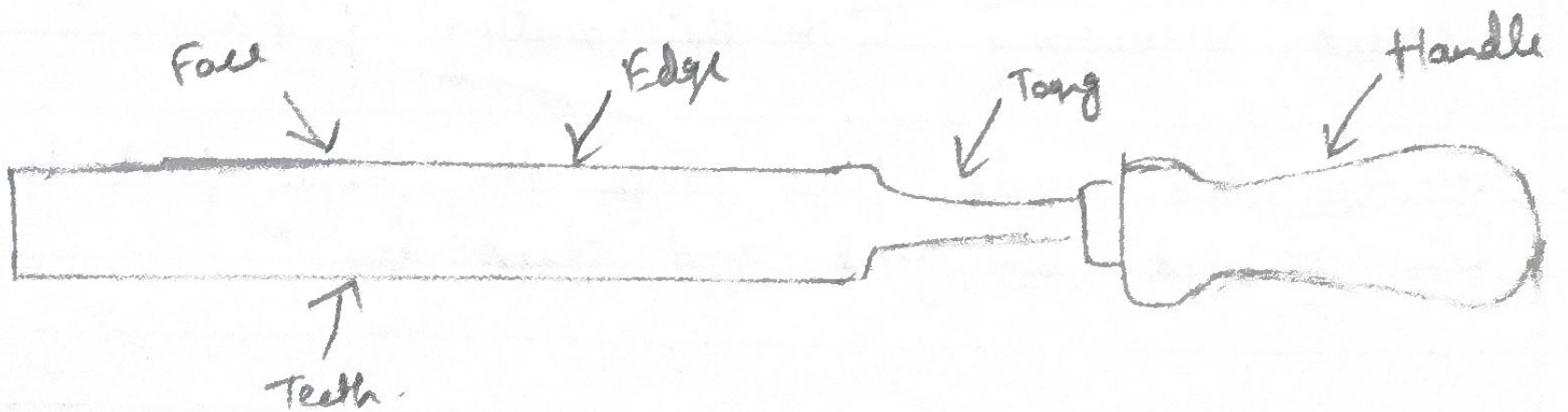
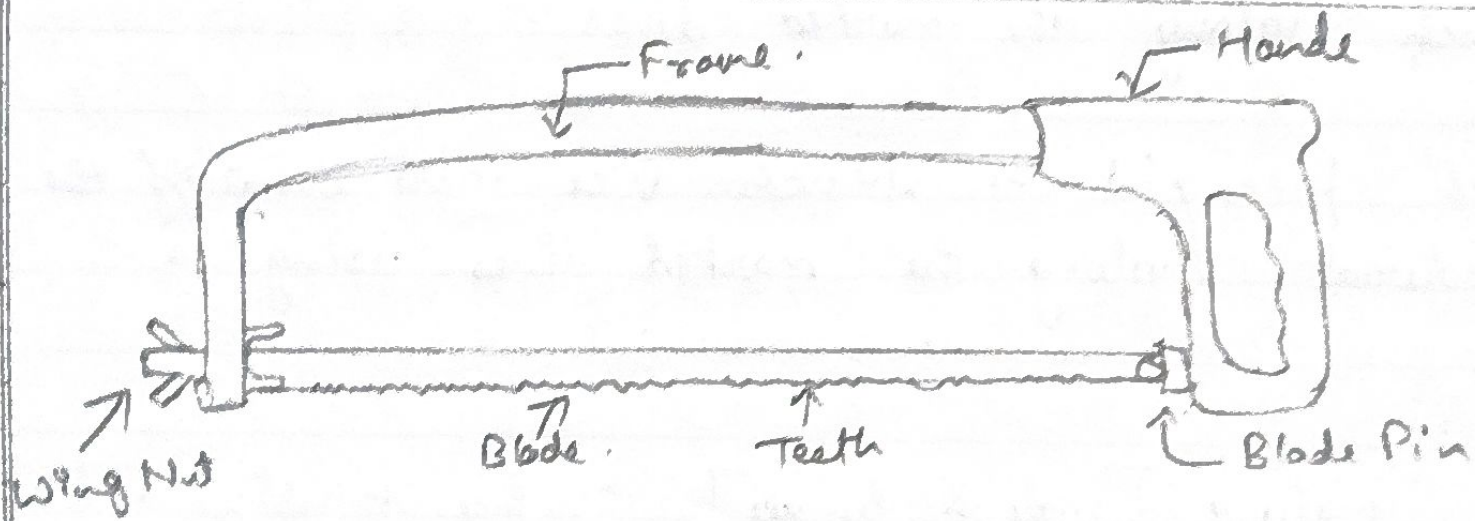
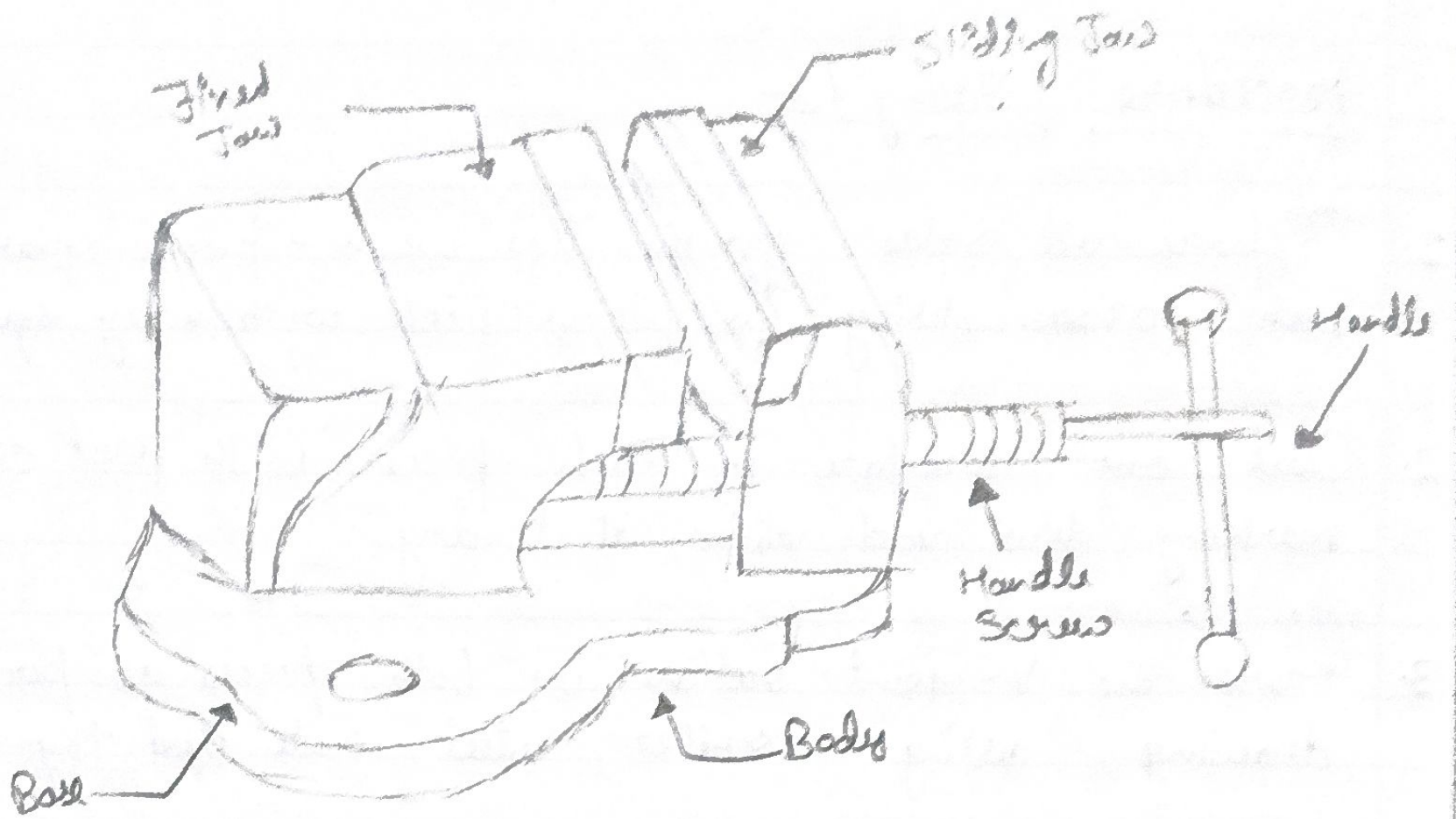


(C) Assembled V-Joint



Procedure [Fitting] -

1. Prepare all sides of the two work-pieces square to each other using a file, check with a try square.
2. Coat one surface of each piece with wet chalk/ marking blue and allow it to dry.
3. Mark the V-joint outline on both pieces as per the drawing, using a scriber, steel rule and try square.
4. Centre-punch along the marked lines.
5. Clamp each piece in the bench vice and saw off the waste material along the marked lines using a hacksaw.
6. File the mating [male & female] surface to size and check accuracy with the mating part.
7. Assemble the male part into the female part to form the finished V-joint and check the fit.



B. Carpentry - Wood Joints -

Carpentry is the process of making wooden components using hand tools, starting from marketable wood and ending in a finished, joined product. Three classical demonstrable/rigid wood joints are studied:

- Cross Half-Lap Joint -

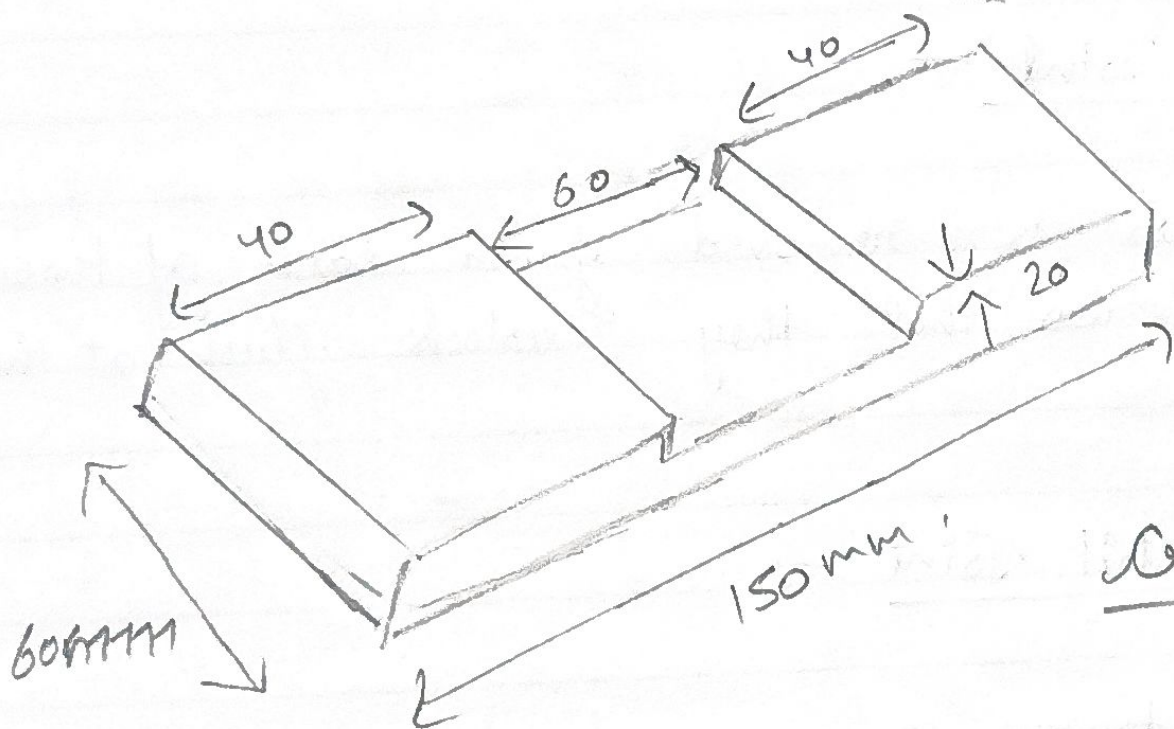
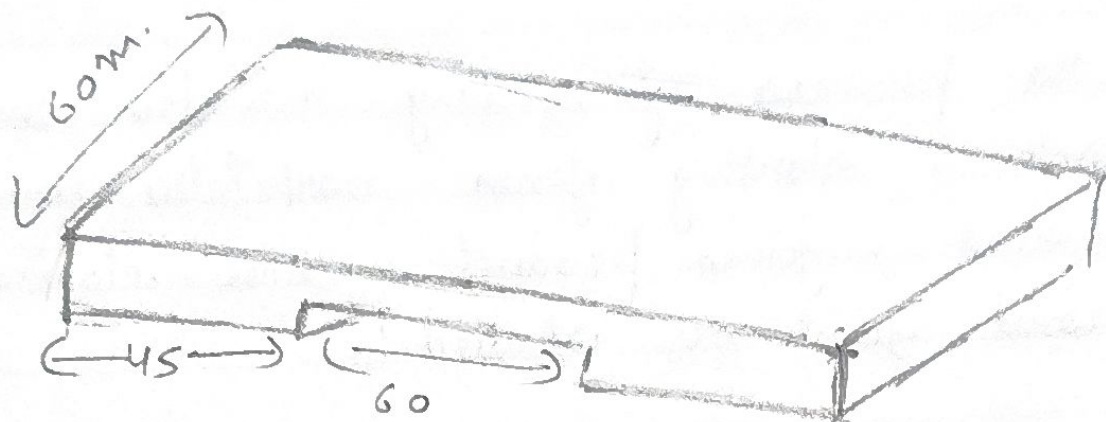
Half the thickness is removed from each of two crossing pieces so that they interlock flush at the crossing point.

- Half-Lap Dovetail Joint -

A trapezoidal tongue on one piece engages a matching recess in the other, resisting pull-apart forces along one direction.

- Mortise & Tenon Joint -

A projecting tenon on one member fits into a matching mortise cut in the other, giving a strong right-angle joint used widely in furniture framing.



Cross Half Lap Joint

Procedure [Carpentry - general, repeated for each joint] -

1. Cut a wooden piece of the required size from the plank using a rip saw.
2. Plane all faces smooth and square using the iron jack plane; Check squareness with a try square.
3. Mark the joints outline on both pieces as per the drawing using a marking gauge, steel rule and pencil.
4. Hold the piece in the carpentry vice and saw along the marked lines.
5. Remove the waste wood using a firmer chisel and mallet, working carefully up to the marked line.
6. Smoothen the worked surfaces with a rasp file and sandpaper.
7. Match and assemble the two pieces to form the required joint and check the fit.

Result —

A male - female V-joint (fitting) and Cross Half-lap Dovetail and Mortise & Tenon wood joints were successfully prepared and assembled as per the given drawings and dimensions.

Precautions —

- Hold the work piece firmly and squarely in the vice before marking, sawing or filing.
- Marking and punching should be done accurately with the work-piece properly supported.
- Always cut on the waste side of the marked line, never on the line itself.
- Use sharp tools; a blunt saw/file is both inaccurate and unsafe.